Business Finance Pro

Version 1.0.4 — Macro-Free Edition

Complete Technical & User Documentation

A self-contained financial operating system for small business owners,
built entirely in Microsoft Excel — no macros, no add-ins, no external services.

This document was first written for v1.0.0 and has been updated for v1.0.4. See section 9.

1. Executive Summary

Purpose

Business Finance Pro replaces the mental overhead of formal accounting with a plain-language, five-minute weekly ritual: log what happened, glance at your health, act on what needs attention. It is a finished product built entirely inside Excel, not a template the customer configures themselves.

Target Users

Freelancers, shop owners, and small service businesses with little or no formal accounting training — owners who currently avoid looking at their numbers because every existing tool assumes either an accountant's vocabulary or hours per week they don't have.

Key Capabilities

- A live Home dashboard summarizing business health, monthly cash flow, urgent action items, and a plain-language weekly narrative
- Guided entry screens for transactions, bills, invoices, and savings/revenue goals
- Three independent rule-based calculation engines plus a shared metrics core — zero AI, zero VBA, formulas only
- A fully validated, protected data layer that prevents accidental corruption while remaining transparent and auditable
- A currency symbol that follows the Currency setting (USD, GBP, EUR, CAD, AUD, PKR)

Design Philosophy

Every design decision was tested against one question: would a non-accountant understand this without help? The workbook avoids accounting terminology, exposes only plain-language design tokens (not raw numbers or jargon) to the presentation layer, and treats calm, non-judgmental tone as a hard requirement. Structurally, it separates data, calculation, and presentation into three strict layers — the same pattern a real SaaS product would use, adapted to what native Excel can do without macros.

2. Feature List

Module	Status	Notes
Home Dashboard	Complete	Business Health, Money In/Out, Needs Attention, Monday Briefing
Settings	Complete	11 editable fields, 4 read-only Product Information fields; the currency symbol follows the Currency setting
Quick Add	Complete	Guided transaction entry with staging workflow
Bills	Complete	Bill list with live Overdue/Upcoming status + add-a-bill workflow; feeds the Attention engine
Clients	Complete	Invoice list with live Overdue/Sent status + add-an-invoice workflow; feeds the Attention engine
Goals	Complete	Goal list with live progress + add-a-goal workflow; feeds Insights
Hidden Data Layer	Complete	9 protected Excel Tables, fully validated
Calculation Engines	Complete	4 independent, modular rule engines
Insights	Complete	Top-3 ranked narrative insights, rule-based
Business Health	Complete	6-state weighted scoring engine, design-token outputs
Needs Attention	Complete	Up to 5 ranked, prioritized action items
Navigation	Complete	Internal links across all 6 visible sheets (fixed in v1.0.2)
Data Validation	Complete	46 rules across 16 reusable dropdown lists
Protection	Complete	All 14 hidden sheets are protected without a password (Review > Unprotect Sheet); visible sheets are locked except inputs
Named Ranges	Complete	108 named ranges, the sole cross-layer interface

3. Workbook Architecture

Sheet Inventory (20 total)

Visible (6): Home, Settings, Quick Add, Bills, Clients, Goals

Hidden (14): _data_Transactions, _data_Categories, _data_Vendors, _data_Bills, _data_Customers, _data_Invoices, _data_Goals, _data_BusinessProfile, _data_Settings, _calc_Core, _calc_Health, _calc_Attention, _calc_Insights, _lists_Validation

Data Flow

User input (visible sheets) -> staging rows -> manual commit (section 6) -> hidden data tables -> _calc_Core (aggregates) -> _calc_Health / _calc_Attention / _calc_Insights -> named ranges -> Home Dashboard.

Calculation Flow

Every visible sheet consumes calculation results exclusively through named ranges — never through direct references to a hidden table or another calculation sheet, with one documented exception: Bills, Clients, and Goals list cards read their own tables directly. No calculation engine references a presentation sheet. This one-directional flow keeps the product debuggable and prevents circular logic.

Overdue status for bills and invoices is not stored: it is calculated from the due date and today's date every time the workbook calculates (changed in v1.0.1).

4. Data Model

Table	Purpose
tbl_Transactions	The atomic record of every dollar moving in or out of the business
tbl_Categories	Plain-language income/expense classification, pre-loaded with 12 defaults
tbl_Vendors	Anyone the business pays
tbl_Customers	Anyone who pays the business
tbl_Bills	Money owed to vendors — one-time or recurring
tbl_Invoices	Money owed by customers
tbl_Goals	Savings, revenue, or debt-payoff targets with live progress tracking
tbl_BusinessProfile	Singleton record of business identity
tbl_Settings	Singleton record of user preferences and product metadata

5. Calculation Engine Overview

_calc_Core

The foundation every other engine builds on. Computes seven month-scoped metrics (Money In, Money Out, Net Cash Flow, Transaction Count, Average Transaction, Current Month, Previous Month) directly from tbl_Transactions using structured references only.

_calc_Health

A four-section, weighted-component scoring engine (Inputs, Intermediate, Threshold Evaluation, Output) producing a plain-language business health state. Restricted by design to consuming only _calc_Core. Currently scores on two real components (Cash Flow, Activity); three components (Cash Runway, Bills, Invoices) exist as architecturally-reserved placeholders.

_calc_Attention

A rule-based engine evaluating up to 7 candidate conditions (5 active, 2 reserved), surfacing up to 5 prioritized, ranked action items. Consumes _calc_Core plus direct structured references to tbl_Bills and tbl_Invoices. Overdue bills and invoices are evaluated live against today's date; Paid bills, and Paid, Void or Draft invoices, are never counted as overdue, and a blank due date is never flagged.

_calc_Insights

A rule-based narrative engine producing the Top 3 "Monday Briefing" insights shown on Home, ranked from 6 candidate rules (5 active). Combines _calc_Core aggregates with category/vendor spend breakdowns and goal progress to write plain-English observations. Amounts in these sentences use the currency symbol from the Currency setting (named range CurrencySymbol).

6. User Guide

Configure Settings

Open Settings, fill in Business Information, Appearance, and Preferences. Currency options are USD, GBP, EUR, CAD, AUD and PKR; changing it changes the symbol shown on screen, not the amounts. The four Product Information fields are system-set and read-only.

Add a Transaction

Go to Quick Add and fill in the fields under "What happened?". A **Ready to save** row builds itself. Until the required fields are filled in, the row stays blank and its heading tells you to fill in the form first. Then follow "Save a record" below.

Add a Bill, a Client / Invoice, or a Goal

Same pattern on the Bills, Clients and Goals screens: fill in the form, then follow "Save a record" below.

Save a record (all screens)

Because the workbook uses no macros, you save the Ready to save row into its data table yourself. It takes about 30 seconds:

1. Fill in the form. Select the **Ready to save** row (first column to last) and press Ctrl+C.
2. Right-click any sheet tab, choose **Unhide**, and pick the data sheet for that screen (table below).
3. On the **Review** tab, click **Unprotect Sheet**. No password is needed.
4. Click the first empty cell in column A, directly under the last row of the table. Right-click, choose **Paste Special**, then **Values**. The table grows by one row.
5. Optional: **Review > Protect Sheet > OK**, then right-click the sheet tab and choose **Hide**.
6. Go back to the Home screen. Your numbers are up to date.

Screen	Data sheet to unhide	Table
Quick Add	_data_Transactions	tbl_Transactions
Bills	_data_Bills	tbl_Bills
Clients	_data_Invoices	tbl_Invoices
Goals	_data_Goals	tbl_Goals

If the table does not grow, check that the sheet is unprotected and that you pasted directly below the last row. Test rows pasted by mistake can be deleted by right-clicking the row in the table and choosing Delete > Table Rows.

Mark a Bill or Invoice as Paid

Overdue and Upcoming (or Sent) status is calculated for you. The only status you set by hand is **Paid** (invoices can also be Void or Draft): unhide _data_Bills or _data_Invoices, unprotect the sheet, and change the Status of that row.

Read the Dashboard

Home shows Business Health, this month's Money In/Out, up to five Needs Attention items, and a weekly narrative under "This Week". The Insights button in the top menu jumps to the "This Week" section.

Understand Business Health

A single plain-language state tells you whether income is comfortably ahead of spending this month. It is intentionally not a number.

Understand Insights

The "This Week" narrative combines up to three automatically-selected observations — biggest expense, top vendor, income, activity, and goal progress.

Understand Needs Attention

Up to five ranked items requiring action: no income logged, spending exceeding income, overdue bills, overdue invoices.

Sample Data

The workbook ships with sample data whose dates are relative to today. Delete the sample rows in each data table before entering your real data.

7. Known Limitations

- **Macro-free staging workflow.** Every add screen requires a manual copy-and-paste-as-values step into a hidden data sheet (section 6) to permanently save an entry — a structural consequence of Excel formulas being unable to write and reset themselves, not fixable without VBA.
- **Currency is a display symbol only.** The symbol follows the Currency setting (USD, GBP, EUR, CAD, AUD, PKR), but amounts are not converted. CAD and AUD use the $ symbol.
- **Placeholder score components.** _calc_Health's Cash Runway, Bills, and Invoice score components are architecturally present but return zero.
- **Placeholder Attention rule.** An "Upcoming loan payment" rule exists as a labeled placeholder; no Loans table exists.
- **Single-goal insight visibility.** Only the highest-priority Active goal is ever mentioned in Insights.
- **List-card architectural exception.** Bills, Clients, and Goals display their lists via direct table references rather than the named-ranges-only pattern used on Home.
- **Three Settings fields have no UI.** InsightsEnabled, CelebrationAnimationsEnabled, and DefaultInvoiceDueDays are real, working fields with no way for a user to change them from any screen.

8. Technical Statistics

Metric	Count
Workbook version	1.0.4 (Macro-Free Edition)
Total sheets	20
Visible sheets	6
Hidden sheets	14
Excel Tables	9
Named ranges	108
Formulas	543
Validation lists	16
Data validation rules	46
Conditional formatting rules	46
Calculation engines	4

9. Version History

The first release was built in strict dependency order across roughly 20 development sprints, organized into five phases:

- **Product definition** — PRD, personas, UX principles, design system, and data model, all locked before construction began.
- **Foundation** — workbook specification, Home and Settings construction, the hidden data layer, and the binding layer connecting them via named ranges.
- **Calculation engines** — _calc_Core, _calc_Health (later refined to a weighted-component architecture), _calc_Attention, and _calc_Insights, each built and tested independently.
- **Remaining modules** — Quick Add, Bills, Clients, and Goals, each activating a previously-placeholder calculation rule as its underlying data became available.
- **Release preparation** — a full RC1 structural audit and an independent adversarial RC2 QA review, both of which found and fixed genuine bugs before the first release.

Releases

Version	Changes
1.0.0	First public release.
1.0.1	Overdue/Upcoming status now calculated live from the due date (it used to freeze once a record was saved). Sample data dates relative to today. Quick Add client lookup fixed; the Vendor / client dropdown includes customers.
1.0.2	All 36 navigation links fixed (they were stored as external file links and showed a security warning). The Insights button jumps to the "This Week" section.
1.0.3	Blank forms no longer produce junk values in the Ready to save rows; dates no longer show #; the Clients due date format corrected; the currency symbol follows the Currency setting; PKR added; product information refreshed.
1.0.4	The 14 hidden sheets no longer need a password to unprotect; the steps for saving a record are documented (README and section 6).

10. Future Roadmap (Not Implemented)

The following require VBA, Office Scripts, or a technology beyond native Excel formulas, and are explicitly out of scope for the macro-free edition:

- One-click data entry, eliminating the manual staging/copy-paste step
- Full Business Health scoring with Cash Runway, Bills, and Invoice components activated
- Multi-goal insight support
- A Loans module (tbl_Loans and associated engine rules)
- Recurring bill/transaction automation
- UI for the three currently-orphaned Settings preference fields
- Currency conversion (the symbol already follows the Currency setting since v1.0.3)
- A companion mobile capture tool that syncs back into this workbook

11. Final Production Assessment

Status: v1.0.4 released. Audited, QA-verified, and corrected after release.

The core financial logic, calculation engines, data layer, and navigation were tested against real recalculation across two independent audit cycles before the first release. Independent testing after release found further problems — overdue status that stopped updating, stale sample data, a client lookup issue, navigation links that did not work, junk values in empty forms, a hardcoded currency symbol, and hidden sheets that needed an undocumented password. All of them were fixed in v1.0.1 to v1.0.4 and re-tested.

The product's central interaction — saving a transaction, bill, invoice, or goal — requires a manual copy-paste step (section 6). This is a deliberate, documented characteristic of the macro-free approach, not an oversight. A future release introducing VBA or Office Scripts could remove that step without altering anything else documented here.